

BACHELOR OF TECHNOLOGY
ELECTRICAL ENGINEERING
THIRD YEAR (FIFTH SEMESTER)
W.E.F. ADMISSION BATCH 2023-24

Sl. No.	Category	Course Code	Course	Contact Hrs. L-T-P	Credit	University Marks	Internal Evaluation
Subject (Theory)							
1	PC	EEPC3001	Control System	3-0-0	3	100	50
2	PC	EEPC3002	Power System - I	3-0-0	3	100	50
3	PC	ECPC3001	Microprocessor & Microcontroller	3-0-0	3	100	50
4	PE	ECPE3001	Electromagnetic Field Theory	3-0-0	3	100	50
		EEPE3002	CAD of Electrical Machines				
		EEPE3003	Energy Conservation & Auditing				
		EEPE3004	Special Electrical Machines				
		-	-				
		-	-				
5	HS	HS3002	Entrepreneurship Development	3-0-0	2	100	50
		HS3003	Professional Ethics				
6	MC	MCMC3001	Environmental Engineering	3-0-0	2	100	50
		MCMC3002	Industrial Safety Engineering				
Subject (Sessional / Practical)							
7	PC	EEPC3201	Control System Laboratory	0-0-3	1.5	-	100
8	PC	EEPC3202	Power System Simulation Laboratory	0-0-3	1.5	-	100
9	PC	EEPC3203	Microprocessor & Microcontroller Laboratory	0-0-3	1.5	-	100
10	PSI	EEPS3201	Seminar on SIRE - I	0-0-3	1.5	-	100
			Total	18-0-12	22	600	700

[Click here to view/download the syllabus of the subjects.](#)

EEPC3001 CONTROL SYSTEM (3-0-0)

Course Objectives :

This course provides fundamental knowledge of control systems, covering mathematical modeling, time and frequency domain analysis, stability criteria (Routh-Hurwitz, Nyquist), and controller design (P, PI, PID). Students will learn state-space representation, system simplification techniques, and apply performance indices to analyze and design effective control systems.

Module-I: (08 Hrs.)

Introduction to Control System: Motivation, Open-loop versus Closed Loop Control System, Examples of Control System, Block Diagram of Control System, Principle of Feedback Control System. Mathematical Modelling and Representation of Systems: Mathematical Modelling of Electrical Systems (RLC Series and Parallel Circuits), Mechanical Systems (Mass-Spring-Damper System, Rotational Mechanical System). Analogous System: Force (Torque)-Voltage Analogy and Force (Torque)-Current Analogy. Introduction to Laplace Transformation and Important Formulas. Transfer Functions: Open-Loop and Closed-Loop Transfer Functions. Block Diagram Algebra: Block Diagram Reduction Techniques, Signal flow graph, Mason's gain formula.

Module-II: (06 Hrs.)

Transient and Steady State Analysis of Linear Time-Invariant (LTI) Systems: Introduction to LTI systems, Standard Test Signals, Time Response of First Order System, Time Response of Second Order System, Time Response Specifications. Steady-State Error and Error Constants. Effect of Adding Zeroes to a System. Performance Indices: ISE, ITSE, IAE, ITAE Indices Comparison.

Module-III: (08 Hrs.)

Frequency Domain Stability Analysis of LTI System: Routh Hurwitz Criteria, Stability Analysis using Root Locus, Stability Analysis using Bode Plot and Nyquist Criteria.

Module-IV: (04 Hrs.)

Controller and Compensator Design: P, PI, and PID Controller Design, Lag, Lead and Lead-Lag Compensator Design.

Module-V: (04 Hrs.)

Analysis of Control Systems in State Space: Introduction, State-Space Representations of Transfer-Function Systems, Solving the Time-Invariant State Space Model, Controllability and Observability.

Course Outcomes:

On completion of this course, students are able to:

- CO-1: Develop the mathematical model of the physical systems and simplify the complicated system.
- CO-2: Employ time domain analysis to predict and diagnose transient performance parameters of the system for standard input functions.
- CO-3: Formulate different types of analysis in the frequency domain to explain the nature of the stability of the system
- CO-4: Identify the needs of different types of controllers and compensators to ascertain the required dynamic response from the system.
- CO-5: Model and Analyse state space representation of LTI system to check the controllability and observability

Text Book(s):

1. K. Ogata, Modern Control Engineering, Pearson Education India, 2015
2. I.J. Nagrath and M. Gopal, Control System Engineering, New age international Publishers, 2009.
3. Norman S. Nise, Control Systems Engineering, Willey India Pvt. Ltd, 2024

Reference Book(s):

1. R.C. Dorf and R.H.Bishop, Modern Control System, Pearson Education, 2017.
2. B.C. Kuo, Automatic Control System, Prentice Hall, 1990.
3. S. Hasan Saeed, Automatic Control Systems (with Matlab Programs), Katson Books, 2017.

EEPC3002 POWER SYSTEM - I (3-0-0)

Course Objectives:

This course provides fundamental knowledge of power systems, covering transmission line parameters, performance analysis, corona effects, and mechanical design of overhead lines. Students will learn to calculate inductance, capacitance, and resistance, analyze distribution systems, and understand underground cables and earthing principles for efficient power system operation.

Module-I: (02 Hrs.)

Introduction to Power System and Generation of Electric Power: Evolution of Power Systems and Present-Day Scenario. Structure of power system. Conventional and non-conventional sources of Electrical Energy.

Module-II: (08 Hrs.)

Transmission Line Parameters: Types of Conductors, Resistance, Inductance of a Conductor due to Internal Flux, Flux Linkages between Two Points External to an Isolated Conductor, Inductance of a Single Phase Two Wire Line, Flux Linkages of one Conductor in a Group, Inductance of Composite-Conductors, Concept of GMD, Transposition of lines, Inductance of a Three Phase Line with symmetrical and Unsymmetrical Spacing, Inductance Calculations for Bundled Conductors, Skin effect and Proximity effect. Capacitance of a Two Wire Line, Capacitance of a Three Phase Line with symmetrical and Unsymmetrical Spacing, Effect of Earth on the Capacitance of a Three Phase Line, Capacitance Calculations for Bundled Conductors, Parallel- Circuit Three Phase Lines.

Module-III: (07 Hrs.)

Performance of Transmission Lines: Representation of Short, medium and long Transmission Line, Equivalent Circuit, Calculation and analysis of performance of transmission lines, Voltage Profile of transmission lines, Ferranti Effect, Power Flow Through Transmission Line, Power Flow capability and Surge Impedance Loading, Series and Shunt Compensation of Transmission Line.

Corona: Theory of corona formation, Electric Stress, Critical Disruptive Voltage, Visual Critical Voltage, Power Loss Due to Corona, Factor affecting corona loss, Methods for reducing corona loss, Advantages and Disadvantages of Corona.

Module-IV: (06 Hrs.)

Mechanical Design of Overhead Transmission Lines: Line Supports, the catenary curve, Sag and Tension Calculation, supports at different levels, stringing chart, Sag template, Conductor Spacing and Ground Clearance, Effects and Prevention of Vibrations (Vibrations and Dampers).

Overhead Line Insulators: Insulator Materials, Types of Insulators, Voltage distribution over a string of suspension insulators, String Efficiency, Methods of improving string efficiency, Testing of insulators.

Module-V: (07 Hrs.)

Distribution Systems: Classification of Distribution Systems, Primary and secondary distribution network, Voltage Drop in DC Distributors, Voltage Drop in AC Distributors, Kelvin's Law, Limitations of Kelvin's Law, Application of Capacitors to Distribution Systems.

Underground Cables: Type and construction, Classification of Cables, Parameters of Single Core Cables, Grading of Cables, Capacitance of Three Core Cable, Comparison of overhead lines with underground Cables.

Power System Earthing: Soil Resistivity, Earth Resistance, Tolerable Step and Touch Voltage, Actual Touch and Step Voltages. Design of Earthing Grid.

Course Outcomes:

On completion of this course, students are able to:

- CO1. Understand the structure and evolution of power systems and distinguish between conventional and non-conventional energy sources.
- CO2. Calculate the resistance, inductance, and capacitance of transmission lines using relevant formulas and techniques.
- CO3. Analyze the performance of different transmission line types (short, medium, and long) and assess voltage regulation and power flow.
- CO4. Evaluate the effects of corona in transmission lines and recommend methods to minimize power loss.
- CO5. Perform mechanical design calculations for overhead transmission lines, including sag, tension, and vibration prevention.

Text Book(s):

- 1. C. L. Wadhwa, Electrical Power Systems, India: New Age International Publishers, 2010.
- 2. A. Hussain, Power System, New Delhi, India: CBS Publishers & Distributors, 1998.
- 3. S. Sivanagaraju, S. Satyanarayana, Electrical Power Transmission and Distribution, Pearson Education, 2009.

Reference Book(s):

- 1. O. I. Elgerd, Electric Energy Systems Theory. New York, NY, USA: McGraw-Hill Education, 1995.
- 2. D. P. Kothari and I. J. Nagrath, Modern Power System Analysis, 4th ed. New Delhi, India: McGraw-Hill Education, 2011.
- 3. R.K. Rajput, Power System Engineering, Laxmi Publications (P) Ltd, 2007.

ECPC3001 MICROPROCESSORS AND MICROCONTROLLERS (3-0-0)

Course Objectives:

This course provides a comprehensive understanding of microprocessor and microcontroller architectures, focusing on the 8086, 8051, and ARM. Students will learn assembly/C programming, peripheral interfacing, and system design. Practical skills include using simulation tools (Keil, Proteus) to develop and debug embedded applications for real-world problem-solving.

Module-I: 8086 Microprocessor Architecture and Programming (07 hours)

Register Organisation of 8086, Architecture of 8086, Pin Configuration of 8086, Memory Organisation, Minimum/Maximum Mode Configuration, Machine Language Instruction Format, Addressing Modes, Instruction Set, Assembler Directives, Assembly Language Programming, Interrupts and its Handling.

Module-II: 8086 Interfacing and Peripherals (08 hours)

8255 PPI: Modes 0, 1, 2 – Programming and Interfacing, 8253 Programmable Interval Timer, 8279 Keyboard/Display Controller, 8259 Programmable Interrupt Controller, Interfacing ADC and DAC, 8251 USART, Stepper Motor Interfacing, Memory Interfacing and I/O-mapped I/O.

Module-III: 8051 Microcontroller Architecture and Programming (06 hours)

8051 Architecture and Pin Configuration, I/O Ports, Memory Organization and Stack, Instruction Set: Data Transfer, Arithmetic, Logical, Branching, and Bit Manipulation, Assembly Programming of 8051, Embedded C Basics For 8051.

Module-IV: 8051 Interfacing and Applications (06 hours)

Programming Timers/Counters, Interrupts in 8051, Interfacing LEDs, Switches, 16x2 LCD, and Sensor Interfacing, Serial Communication.

Module 5: Introduction to ARM Microcontrollers (3 hours)

RISC vs CISC concepts, Basics of ARM Architecture and Instruction Set, Pipelining concept.

Course Outcomes:

By the end of this course, students will be able to:

- CO1: Apply foundational knowledge of electronics, digital systems, and embedded systems to analyze microprocessor/microcontroller-based problems.
- CO2: Identify and analyze engineering problems in processor and peripheral interfacing, and propose effective solutions.
- CO3: Design embedded system solutions using microcontrollers for real-world applications, considering functional and practical constraints.
- CO4: Conduct experiments with microcontrollers, evaluate outputs, and interpret results for debugging and performance analysis.
- CO5: Use software tools (e.g., Keil, Proteus, MPLAB, STM32Cube IDE) and hardware kits for simulation, testing, and development of microcontroller-based applications.

Text Books:

1. Advanced Microprocessors and Peripherals, A.K. Ray, K M Bhurchandi, TMH Publication, 2007.
2. The 8051 Microcontroller and Embedded Systems, Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D.M C Kinlay, Pearson Education, Second Edition, 2008.
3. ARM System Developer's Guide – Sloss, Symes and Wright, Published by ELSEVIER INDIA, 2014.

References:

1. Microprocessors and Interfacing, Programming and Hardware, Douglas V Hall, TMH Publication, 2006.
2. Microcontrollers: Principles and Application, Ajit Pal, PHI Publication.
3. The Definitive Guide To ARM CORTEX M3 And CORTEX M4 Processors by Joseph Yiu, Newnes Publication, 3rd edition.

ECPE3001 ELECTROMAGNETIC FIELD THEORY (3-0-0)

Course Objectives

This course introduces fundamental concepts of electromagnetic fields, covering coordinate systems, vector calculus, electrostatic and magnetostatic fields, Maxwell's equations, and wave propagation. Students will learn to analyze transmission lines, apply boundary conditions, and solve field-related problems using theoretical principles and mathematical formulations.

Module-I: (06 Hrs.)

Co-ordinate systems & Transformation: Cartesian co-ordinates, circular cylindrical co-ordinates, spherical co-ordinates.

Vector Calculus: Differential length, Area & volume, Line surface and volume Integrals, Del operator, Gradient of a scalar, Divergence of a vector & divergence theorem, curl of a vector & Stoke's theorem, Laplacian of a scalar.

Module-II: (08 Hrs.)

Electrostatic Fields: Coulomb's Law, Electric Field Intensity, Electric Fields due to point, line, surface and volume charge, Electric Flux Density, Gauss's Law – Maxwell's Equation, Applications of Gauss's Law, Electric Potential, Relationship between Electric Potential (E) and Potential Gradient (V) –Maxwell's Equation an Electric Dipole & Flux Lines, Energy Density in Electrostatic Fields.

Electrostatic Boundary – Value Problems: Poisson's & Laplace's Equations, Uniqueness theorem, General procedures for solving Poisson's or Laplace's Equation.

Module-III: (08 Hrs.)

Magnetostatic Fields: Biot-Savart's Law, Magnetic Field intensity due to a finite and infinite wire carrying current; Magnetic field intensity on the axis of a circular and rectangular loop carrying current; Magnetic flux Density; Magnetic Vector Potential; Ampere's circuital law and simple applications; Inductance and mutual inductance. Boundary conditions for electric fields and magnetic fields. Conduction current and displacement current densities; Continuity equation for current; Maxwell's Equation in Differential and Integral form from Modified form of Ampere's circuital law, Faraday's Law and Gauss's Law.

Module-IV: (04 Hrs.)

Electromagnetic Fields and Wave Propagation: Faraday's Law, Transformer & Motional Electromagnetic Forces, Displacement Current, Maxwell's Equation in Final forms, Time Varying Potentials, Time-Harmonic Field.

Electromagnetic Wave Propagation: Wave Propagation in lossy Dielectrics, Plane Waves in loss less Dielectrics, Power & pointing vector.

Module-V: (04 Hrs.)

Transmission line: Waves in transmission line, Line parameters, Transmission line equation & solutions, Propagation constants, Characteristic impedance, Wavelength, Velocity of propagation. Standing Wave Ratio (SWR), impedance matching. Solution of problems. Electromagnetic interference.

Course Outcomes:

On completion of this course, students are able to:

- CO-1: Understand and apply coordinate systems and vector calculus for formulating electromagnetic field problems.
- CO-2: Analyse electrostatic fields and apply Gauss's law, electric potential, and energy density in solving field-related problems.
- CO-3: Apply magnetostatic laws and principles (Biot-Savart, Ampere's) to compute magnetic fields and understand inductance.

MCMC3002 INDUSTRIAL SAFETY ENGINEERING (3-0-0)

Course Objectives:

1. Students will be able to recognize and evaluate occupational safety and health hazards in the workplace, and to determine appropriate hazard controls following the hierarchy of controls.
2. Students will furthermore be able to analyze the effects of workplace exposures, injuries and illnesses, fatalities and the methods to prevent incidents using the hierarchy of controls, effective safety and health management systems and task-oriented training.

Course Outcomes:

By the end of this course, a student should:

- CO1: Evaluate workplace to determine the existence of occupational safety and health hazards
- CO2: Identify relevant regulatory and national consensus standards along with best practices that are applicable.
- CO3: Select appropriate control methodologies based on the hierarchy of controls
- CO4: Analyze injury and illness data for trends

Module-I: (07 hrs)

Industrial safety: Accident, causes, types, results and control, mechanical and electrical hazards, types, causes and preventive steps/procedure, describe salient points of factories act 1948 for health and safety, wash rooms, drinking water layouts, light, cleanliness, fire, guarding, pressure vessels, etc, Safety color codes. Fire prevention and firefighting, equipment and methods.

Module-II: (07 hrs)

Fundamentals of maintenance engineering: Definition and aim of maintenance engineering, Primary and secondary functions and responsibility of maintenance department, Types of maintenance, Types and applications of tools used for maintenance, Maintenance cost & its relation with replacement economy, Service life of equipment.

Module-III: (07 hrs)

Wear and Corrosion and their prevention: Wear- types, causes, effects, wear reduction methods, lubricants-types and applications, Lubrication methods, general sketch, working and applications, i. Screw down grease cup, ii. Pressure grease gun, iii. Splash lubrication, iv. Gravity lubrication, v. Wick feed lubrication vi. Side feed lubrication, vii. Ring lubrication, Definition, principle and factors affecting the corrosion. Types of corrosion, corrosion prevention methods.

Module-IV: (07 hrs)

Fault tracing: Fault tracing-concept and importance, decision tree concept, need and applications, sequence of fault-finding activities, show as decision tree, draw decision tree for problems in machine tools, hydraulic, pneumatic, automotive, thermal and electrical equipment's like, i. Any one machine tool, ii. Pump iii. Air compressor, iv. Internal combustion engine, v. Boiler, vi. Electrical motors, Types of faults in machine tools and their general causes.

Module-V: (08 hrs)

Periodic and preventive maintenance: Periodic inspection-concept and need, degreasing, cleaning and repairing schemes, overhauling of mechanical components, overhauling of electrical motor, common troubles and remedies of electric motor, repair complexities and its use, definition, need, steps and advantages of preventive maintenance. Steps/procedure for periodic and preventive maintenance of: i. Machine tools, ii. Pumps, iii. Air compressors, iv. Diesel generating (DG) sets, Program and schedule of preventive maintenance of mechanical and electrical equipment, advantages of preventive maintenance. Repair cycle concept and importance.

Books:

1. Maintenance Engineering Handbook, Higgins & Morrow, Da Information Services.
2. Maintenance Engineering, H. P. Garg, S. Chand and Company.
3. Pump-hydraulic Compressors, Audels, McGraw Hill Publication.
4. Foundation Engineering Handbook, Winterkorn, Hans, Chapman & Hall London.

SHHS3002 ENTREPRENEURSHIP DEVELOPMENT (3-0-0)

Course Objectives –

1. To explain concept of entrepreneurship and build and understanding about business situation in which entrepreneurs act.
2. To explain classification and type of entrepreneurs and the process of entrepreneurial project development
3. To discuss the steps in venture development and new trends in entrepreneurship.
4. The more focus is given on creativity and innovation.

Module-I: (10 hours)

Entrepreneurship: Concept of entrepreneurship and intrapreneurship, Types of Entrepreneurs, Nature and Importance, Entrepreneurial Traits and Skills, Entrepreneurial Motivation and Achievement, Entrepreneurial Personality

Module-II: (08 hours)

Entrepreneurial Environment, Identification of Opportunities, Converting Business Opportunities into reality. Start-ups and business incubation, Setting up a Small Enterprise. Issues relating to location, Environmental Problems and Environmental pollution Act, Industrial Policies and Regulations

Module-III: (10 hours)

Need to know about Accounting, Working capital Management, Marketing Management, Human Resources Management, and Labour Laws. Organizational support services - Central and State Government, Incentives and Subsidies.

Module-IV: (12 hours)

Sickness of Small-Scale Industries, Causes and symptoms of sickness, cures of sickness, Role of Banks and Governments in reviving industries.

Course Outcomes

After completion of this course, students

- CO1: will aware about foundation of entrepreneurship development and its theories
CO2: will identify the type of entrepreneur and the steps involved in a entrepreneurial venture.
CO3: will understand various steps involved in starting a venture and to explore marketing methods & new trends in entrepreneurship.
CO4: Think creative and innovative

Books:

1. Entrepreneurship Development and Management, Vasant Desai, HPH
2. Entrepreneurship Management, Bholanath Dutta, Excel Books
3. Entrepreneurial Development, Sangeeta Sharma, PHI
4. Entrepreneurship, Rajeev Roy, Oxford University Press

EEPC3201 CONTROL SYSTEM LABORATORY (0-0-3)

Course Objectives:

This lab course provides hands-on experience in analyzing and designing control systems. Students will study DC and AC motor control, design compensators, implement PID controllers, validate transfer functions, and use MATLAB for stability analysis (root locus, Bode plots) to evaluate system performance.

1. Study of a dc motor driven position control system. (03 Hrs)
2. Study of speed torque characteristics of two-phase AC servomotor and determination of its transfer function. (03 Hrs)
3. Obtain the frequency response of a lag and lead compensator. (03 Hrs)
4. To observe the time response of a second order process with P, PI and PID control and apply PID control to servomotor. (03 Hrs)
5. To determine the transfer function of a system (network) using transfer function analyser. (03 Hrs)
6. To study and validate the controllers for a temperature control system. (03 Hrs)
7. To study the position control system using Synchroscope. (03 Hrs)
8. To Analyse the Time Domain specifications of Under damped second order system using MATLAB. (03 Hrs)
9. To analyse the stability of the system by using Root locus using MATLAB. (03 Hrs)
10. To analyse the stability of the given linear system using Bode plot using MATLAB. (03 Hrs)

Course Outcomes:

- CO1: Analyse and assess the position control system of DC motors to evaluate system performance.
- CO2: Investigate the speed-torque characteristics of a two-phase AC servomotor and derive its transfer function.
- CO3: Design and evaluate the frequency response of lag and lead compensators for performance enhancement.
- CO4: Observe and analyse the time response of second-order processes using P, PI, and PID controllers and implement PID control for a servomotor.
- CO5: Determine and validate the transfer function of a system using a transfer function analyser.

EEPC3202 POWER SYSTEM SIMULATION LABORATORY (0-0-3)

Course Objectives

This lab course provides hands-on experience in power system simulation using MATLAB/SIMULINK. Students will analyze transmission line parameters, model short/medium/long lines, study Ferranti effect, calculate sag and string efficiency, improve voltage profiles, and compare distribution systems to develop practical skills in power system analysis and design.

1. Determination of transmission line parameters (inductance and capacitance) for a single-phase line using MATLAB Program / SIMULINK Model. (3Hrs)
2. Determination of transmission line parameters (inductance and capacitance) for a symmetrical & unsymmetrical three-phase transmission line using MATLAB Program / SIMULINK Model (3Hrs)
3. Determination of Power Angle Diagrams, Reluctance Power, Excitation, Emf and Regulation for Salient and Non-Salient Pole Synchronous Machines using MATLAB Program / SIMULINK Model (3Hrs)
4. Modelling and performance analysis of short & medium transmission line using MATLAB Program / SIMULINK Model (3Hrs)
5. Study of Surge Impedance Loading of a long transmission line using MATLAB Program / SIMULINK Model (3Hrs)
6. Verify the Ferranti effect in a transmission line using MATLAB Program / SIMULINK Model (3Hrs)
7. Evaluation of sag in overhead transmission lines with equal and unequal tower heights using MATLAB Program / SIMULINK Model (3Hrs)
8. Determination of string efficiency of suspension-type insulators using MATLAB Program / SIMULINK Model (3Hrs)
9. Improvement of voltage profile at a load bus using a shunt capacitor using MATLAB Program / SIMULINK Model (3Hrs)
10. Study and compare the performance of a ring and radial distributions systems using MATLAB Program / SIMULINK Model (3Hrs)

Course Outcomes:

- CO1: Analyse and calculate the inductance and capacitance of single-phase and three-phase transmission lines (symmetrical and unsymmetrical configurations).
- CO2: Develop models for short, medium, and long transmission lines and evaluate their performance parameters such as voltage regulation and efficiency.
- CO3: Interpret and analyse the Ferranti effect in medium and long transmission lines using appropriate simulation or experimental techniques.
- CO4: Compute the sag and tension in overhead transmission lines for both equal and unequal tower heights and string efficiency of suspension-type insulators.
- CO5: Analyse the voltage profile at a load bus using a shunt capacitor and compare the performance of a ring and radial distributions systems .

EEPC3203 MICROPROCESSOR AND MICROCONTROLLER LABORATORY (0-0-3)

Course Objectives:

This laboratory course aims to develop practical skills in microprocessor (8086) and microcontroller (8051/Arduino) programming and interfacing. Students will implement arithmetic, sorting, and communication programs, interface peripherals (ADC/DAC, motors), and design embedded systems (sensors, IoT applications). Emphasis is placed on hardware-software integration, debugging, and using development tools for real-world problem-solving.

(Any Ten of the following experiments are to be performed)

1. Programs for 16-bit arithmetic operations using 8086.
2. Programs for Sorting and Searching (Using 8086).
3. Programs for String manipulation operations (Using 8086).
4. Programs for Digital clock and Stop watch (Using 8086).
5. Interfacing ADC and DAC with 8086.
6. Parallel Communication between two Micro Processor Kits using Mode 1 and Mode 2 of 8255.
7. Interfacing and Programming 8279, 8259, and 8253 with 8086.
8. Serial Communication between two Micro Processor Kits using 8251.
9. Interfacing and Programming of Stepper Motor and DC Motor Speed control with 8086.
10. Programming using Arithmetic, Logical and Bit Manipulation instructions of 8051 microcontroller.
11. Programming and verifying Timer, Interrupts and UART operations in 8051
12. A design problem using 8051 (A problem like multi-parameter data acquisition system, elevator simulation, traffic simulation, digital clock using LED matrix, etc)
13. To measure the distance of the obstacle using ultrasonic sensor and Arduino and display the distance on the LCD/Graphical LCD.
14. To read different parameters like temperature, humidity, motion etc. interfaced with Arduino uno/any microcontroller and send these data to the server (like Raspberry Pi) through wireless communication interfaces like Bluetooth, Wi-Fi, ZigBee, etc and store in the database for analytics.
15. To design/simulate a security system using RFID module and alert the authorities via SMS (GSM module) with geocoordinates (GPS module) during unauthorised access. Course

Outcomes:

Upon completion of the subject, the students will demonstrate the ability to:

- CO1: Apply theoretical knowledge of microprocessors and microcontrollers to develop low-level code and interface with external devices.
- CO2: Identify and analyse interfacing problems and debugging issues during hardware-software integration.
- CO3: Design functional embedded systems like clocks, control systems, and smart sensors using Microprocessor/Microcontroller Units and analyse performance parameters such as timing, delay, and communication accuracy during system testing.
- CO4: Use development tools like Keil, Proteus, STM32CubeIDE, and MPLAB to simulate, debug, and test embedded systems.
- CO5: Adapt multiple microcontroller platforms, communication standards, and development environments.